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Application of Polarography

The Polarography is used in many field of life and also in many discipline. It can be used for both Qualitative and Quantitative Analysis.

Applications:-

The application of Polarography are following.

Use to study

- Ultra trace range metal (sub ppb)
- Waste water Analysis
- Plating Analysis
- Biochemical Analysis
- Environmental Analysis.
- Liquor Analysis

The major applications are following

- Pharmaceutical Analysis
- Organic Analysis
- Inorganic Analysis
- Industrial Applications

Industrial Application

The polarography has many industrial application like it is helpful in food industry and many other industries.

- In food industry it is used for checking the purity of compounds like honey, sugar etc.

- It can be used for determination of Ascorbic acid (**Vitamin C**) in many Citrus juice by standard addition and Calibration curve method.

- The Polarography can be used to determined the water content by using Karl Fisher titration. So we can determined that it may be pure or not etc.

The water can be determined by following method.

1ml of KF reagent = 6.66 mg water

Vol. of KF added = 2.0 ml.

Excess of KF reagent = 1.18 ml of H_2O

Titre of KF = 1.46 ml.

$$\text{water content} = \frac{1.46 \times 6.6 \times 100}{10.0 \times 1000} = 0.098\% \text{ (w/v)}$$

Inorganic Application:-

• Monovalent, bivalent and trivalent ions can be reduced on the surface of D.M.E. So can be analyzed by polarography.

• The amount of **Cu, Cr, Ni, Mo etc** present in the steel sample can be determined by polarography.

The analysis of anions as chromate, iodate can be done by this method.

• The Zn and many other metal is determined.
Example:-

Zinc and Nickel in pure copper salt can be dissolved in ammoniacal solution (NH_4Cl) and electrolyzed at about 0.7 volt VS SCE. Then the determined zinc and nickel as less as 0.00001% nickel can be determined.

• Many inorganic ions can be identified by using polarography like

Cl^- , ClO^- , ClO_2^- , ClO_3^- , Cl_2 , Br_2 , I , I_2 , SO_3^{2-} , SCN^- , $\text{S}_2\text{O}_5^{2-}$ etc.

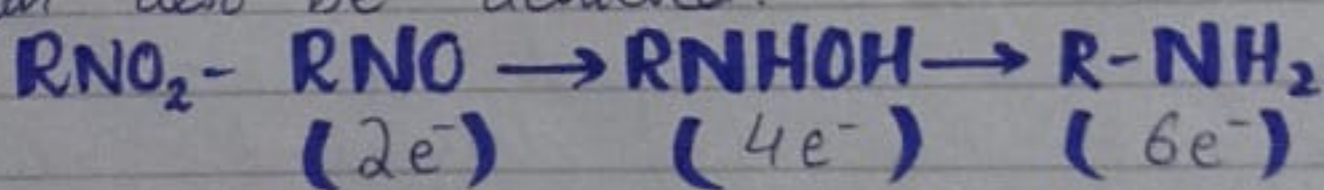
• The amount of titanium present in Al-alloy can also be determined.

Organic Analysis:-

The Organic Analysis include analysis of Carbonyl, peroxide and many other functional group like

$C=C$, $C-X$, $C=O$, $C=N$, $N=N$, $N=O$, NO_2 and these all are electroactive organic functional group. It is possible to analyze simultaneously a mixture of 3 or 4 electroactive ions. and functional group is inferred from half wave potential.

• Multistage reduction of group like nitro to nitroso to hydroxyl amine to amino group can also be achieved.



• To obtained a wave on polarographic curve, the organic substance should undergo either an electro-reduction giving an cathodic wave or an oxidation giving an anodic wave under the galvanometer zero-line.

• Organic compounds that are reduce at DME must contain a strongly polar or an unsaturated bond.

Vitamins:-

Most water-soluble and some oil soluble vitamins are either oxidized or reduced at the dropping mercury electrode and can be estimated in pure solution. Unfortunately most of the assays are unsuitable for food and natural products. Methods are useful for

Aneurine, Riboflavine, nicotinic acid, pantothenic acid, pyridoxine, folic acid, vitamin B₁₂ and vitamin K.

Hormones:-

Thyroxine, insulin, adrenaline and several other hormones give characteristic polarograms, that under certain conditions can be used for their determination.

There are many other such applications of Polarogram in Pharmaceutics.

Qualitative Analysis:-

The qualitative analysis includes many methods for analysis.

The qualitative analysis includes both organic and inorganic analysis.

Trace Metals and Metal Containing Drugs

The metal examined includes antimony, Arsenic, cadmium, Vanadium, Mercury, Iron, Lead, Magnesium etc.

Blood and Urine contain traces of arsenic and Vanadium which needs lengthy pretreatments to remove reducible organic matter before an extract suitable for polarography is obtained.

Many scientist like Page, Robinson, Codex performs for polarography application by using milk, Gelatin and other materials.

Antiseptics and Insecticides

Several mercury containing antiseptics and new insecticides gammexane and parathion, can be determined polarographically.

Example:-

Page and Waller have shown that thiomersalate in ~~1.0~~ 1.0N hydrochloric acid give well-defined steps at relatively low potential $-0.48V$ and are suitable for routine determination of thiomersalate and many other pharmaceutical vaccines.

Pharmaceutical Application :-

The pharmaceutical Application includes

- Dissolved Oxygen and peroxides
- Trace metal and metal-containing drugs
- Antiseptics and insecticides
- Vitamins
- Hormones
- Antibodies
- Alkaloids
- Blood serum and cancer diagnosis

Dissolved Oxygen and Peroxides :-

Oxygen dissolved in an electrolyte solution is reduced at mercury electrode (DME) and yield two distinct step.

(1) The first step, at $-0.1V$, is due to the reduction of oxygen to hydrogen peroxide

(2) The second, at $-0.9V$, is due to reduction of hydrogen peroxide to either water or hydroxyl ion. These steps used for determination of oxygen content in blood, blood body fluids, milk and in liquor.